

Akash Karthik

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EDUCATION

Bachelor of Science, Computer Science

Expected May 2027

San Jose State University, San Jose, CA

Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Discrete Math, Computer Systems, Intro to Circuits

TECHNICAL SKILLS

Languages: C, C++, Python, Assembly, Java

Algorithms & Sensors: Fixed-point PI/PID, state estimation, visual-inertial and IMU sensor fusion, anti-windup, slew-rate limiting, smoothing and debounce logic

Embedded: STM32H7, TTC 60 VCU, PX4, real-time firmware, CAN, SPI, microcontroller I/O

Python & Tools: NumPy, pandas, Matplotlib, OpenCV, pytest, software-in-the-loop replay, TASKING, BusMaster, MoTeC i2 Pro, GDB, Git

EXPERIENCE

Software Engineer Intern

Feb 2026 - Present

Argus Defense | San Jose, CA

- Integrated and tuned visual-inertial odometry for onboard state estimation, fusing optical and inertial sensor data to improve position-hold accuracy
- Developed and validated embedded flight-control firmware in C on PX4, iterating control gains against logged flight telemetry
- Tuned PID flight-control loops on a PX4 flight controller to reduce vertical and horizontal position drift in autonomous flight

Software Lead

Jun 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

- Lead software planning for VCU controls, embedded firmware, telemetry, CAN diagnostics, and validation workflows
- Coordinate priorities and technical reviews across power limiting, launch control, dashboard, simulation, and vehicle testing

Software Controls Engineer

Jul 2025 - Jun 2026

Spartan Racing, Formula SAE | San Jose, CA

- Implemented a torque-to-power algorithm in C with fixed-point PI control, saturation, anti-windup, and slew limits for deterministic TTC 60 VCU execution
- Designed a lap-adaptive energy-management algorithm setting a power setpoint from cumulative energy versus budget, holding actual power within **0.01%** of setpoint
- Built CAN diagnostics firmware for live power and torque frames, plus a software-in-the-loop replay harness to verify controller changes against recorded telemetry

Software Intern

Aug 2024 - Jun 2025

Spartan Racing, Formula SAE | San Jose, CA

- Built deterministic entry and exit state logic for power limiting from measured power, requested torque, and threshold flags
 - Tuned the PI torque controller used for power limiting to within **2%** error, smoothing torque transitions to preserve drivability
 - **Results:** 1st in endurance at MIS EV 2025
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PROJECT EXPERIENCE

Posture Detection System

2025

Python, OpenCV, real-time vision, sensor telemetry

- Developed a real-time pose-detection pipeline classifying posture from camera data with temporal smoothing, debounce logic, and configurable alert thresholds
- Instrumented FPS and latency, recorded sessions, and exported labeled frames and CSV metrics for offline analysis

Formula SAE EV Dashboard

Feb 2026 - Present

Spartan Racing, Formula SAE | San Jose, CA

C, STM32H7, CAN, SPI

- Built a real-time CAN telemetry display on an STM32H7 driving an LCD over SPI, decoding HV voltage, throttle, cell temperature, and power budget
- Designed an energy bar and LED alert system giving the driver lap-by-lap feedback on energy budget